Optimal Execution of Large NSE Equity Orders using Deep Reinforcement Learning

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Code, trained agents & results: `https://github.com/Harsh-nandanshukla/rl-optimal-execution`

Abstract. A fund that must sell a large block of shares within one trading day faces a trade-off: selling fast moves the price against it (market impact), selling slowly exposes it to price risk. We cast this as a finite-horizon MDP and train DQN (written from scratch) and PPO agents to choose how much to sell in each of 25 fifteen-minute intervals. Because the environment's return equals the Almgren–Chriss (AC) objective exactly, we can test RL against a known optimum. On 10,000 held-out simulated days, both agents recover the AC schedule to within 0.14–0.17 bps (98% of AC's gain over TWAP). When prices carry a predictable short-term drift, the agents beat AC by 14–15 bps and a hand-written signal rule by 4–5 bps; hiding the signal removes the edge. On real NSE minute data (2023–26), the signal-free agents match AC and beat TWAP by 8 bps, while signal-trained agents do not transfer.

1 Problem and Motivation

Institutional orders are far larger than the liquidity available at any instant, so they are split into child orders over the day. Each child order pays a *temporary impact* cost that grows with trading speed, and every sale also lowers the price permanently. Trading slowly avoids impact but leaves inventory exposed to price volatility. Almgren and Chriss [1] solved this trade-off in closed form under a random-walk price; industry benchmarks such as TWAP and VWAP ignore risk entirely. Our question is whether a learned policy can (i) rediscover the analytical optimum when it exists, and (ii) beat it when prices are partly predictable, which the static AC schedule cannot exploit. RL fits because decisions are sequential and the cost is only known at the end of the day.

2 RL Formulation (MDP)

Environment. Units are normalised: arrival price $S_0{=}1$, order $X{=}1$; costs $\times 10^4$ are bps. A day has $N{=}25$ decisions, $\tau{=}1/25$. At step k the agent sells n_k, leaving inventory $x_k = x_{k-1} - n_k$ ($x_0{=}1$). Discrete-time Almgren–Chriss dynamics [1]:

$$\tilde{S}_k = S_{k-1} - \varepsilon - \eta\,\frac{n_k}{\tau}, \qquad S_k = S_{k-1} + \sigma\sqrt{\tau}\,\xi_k + \mu_k\tau - \gamma n_k,$$

with execution price $\tilde{S}_k$ (spread $\varepsilon{=}0.5$ bps, temporary impact $\eta{=}0.002$), permanent impact $\gamma{=}0.001$, $\sigma{=}2\%$/day, $\xi_k \sim \mathcal{N}(0,1)$. Drift $\mu_k{=}0$ in Exp A; AR(1) in Exp B ($\mu_k{=}\phi\,\mu_{k-1}{+}$noise, $\phi{=}0.8$, s.d. 0.04). On real NSE days the increments come from data.

State ($\mathscr{S} \subset \mathbb{R}^4$, continuous): $s_k{=}\big(x_k, \frac{N-k}{N}, \frac{\mu_k}{0.04}, \frac{\Delta S_k}{\sigma\sqrt{\tau}}\big)$: inventory left $\in[0,1]$, time left $\in[0,1]$, normalised observed drift and last price move, clipped to $[-5,5]$. The ablation sets the third entry to 0.

Action ($\mathscr{A}$, 12 discrete): action a sells $n_k = m_a\,x_{k-1}/(N{-}k{+}1)$, $m_a \in \{0, 0.25, \ldots, 2, 2.5, 3, 4\}$, so $m{=}1$ is "TWAP pace from here", <1 slows down, >1 speeds up. Any inventory left at the final step is force-sold.

Reward (per step, bps):

$$r_k = 10^4\big[\underbrace{n_k(\tilde{S}_k - S_{k-1})}_{\text{impact+spread}} + \underbrace{x_k(S_k - S_{k-1})}_{\text{inventory P\&L}} - \underbrace{\lambda\sigma^2\tau x_k^2}_{\text{risk}}\big].$$

The first term charges for selling fast, the second marks unsold shares to market, the third discourages holding inventory ($\lambda{=}25$). The P&L terms telescope, so with $\gamma_{\text{RL}}{=}1$ the return is exactly

$$\textstyle\sum_k r_k = -\big(\text{IS} + \lambda\sigma^2\tau\sum_k x_k^2\big), \quad \text{IS} = 1 - \sum_k n_k\tilde{S}_k,$$

minus the AC mean–variance objective (verified to 10^{-12} in code). A cost-only reward was rejected: it optimises a different objective and could never match AC. In a random walk the optimal policy is therefore the AC schedule $x_k^\star = \sinh(\kappa(T{-}t_k))/\sinh(\kappa T)$ ($\kappa T \approx 2.25$), giving Exp A a known right answer.

3 Method

Algorithms. The action set is small and discrete, so we use a value-based learner (DQN) and, for comparison, a policy-gradient one (PPO). **DQN [2] (from scratch, PyTorch).** MLP $4{\to}128{\to}128{\to}12$, replay buffer 100k, Double-DQN targets, soft target update ($\tau{=}0.005$), Huber loss, Adam (lr $5{\times}10^{-4}$), gradient clipping, ε decayed $1{\to}0.05$ over 20k steps. **PPO** [3] (Stable-Baselines3 [4], permitted by the TAs) uses the same 12 actions, a $128{\times}128$ network and $\gamma{=}1$; hyperparameters were fixed in advance, not tuned. **Variance-reduced training reward.** Price noise ($\sim$100 bps/day) swamps the 7.6 bps AC–TWAP gap, so on synthetic paths we train on the conditional expectation of the noise term. This is unbiased under $\gamma{=}1$ and leaves the optimal policy unchanged; all evaluation uses realised costs. **Protocol.** Each configuration is trained for 300k steps (12,000 days) with 5 seeds on CPU. The best checkpoint is chosen on 1,000 fixed validation paths. Models are then committed and frozen with a SHA-256 manifest, and the held-out test set (10,000 paths) is evaluated *once*. All strategies see identical paths, so we report paired differences with 95% CIs, plus a t-interval across seeds. Baselines: Immediate, TWAP, VWAP, closed-form AC, and AC snapped to the 12-action grid ("AC on grid"). **Real data.** 1-minute NSE bars for ADANIPORTS, BAJAJ-AUTO and COALINDIA (Kaggle, 2015–26) are cut into clock-time 15-min buckets, keeping full sessions only. Split: train $\leq$2021, validation 2022, test 2023-01 to 2026-04 (803 dates, 2,409 stock-days). Impact is still simulated on the real price paths; VWAP uses the training-period volume profile (no look-ahead), and CIs are clustered by date because the three stocks share dates.

4 Results

Experiment A: RL recovers the optimum. In a pure random walk both agents land within 0.14 (DQN) and 0.17 bps (PPO) of AC on 10,000 test days, keeping 98.1% and 97.7% of AC's 7.6 bps advantage over TWAP (Table 1). Their mean inventory curves are visually indistinguishable from AC (Fig. 1a): holdings stay within about 1–2% of the order at every step, and the error is almost entirely a fixed bias with near-zero path-to-path dispersion. The agents therefore learned a static, AC-shaped schedule and correctly ignore price noise, which is exactly what theory prescribes here. 9 of 10 agents beat AC snapped to the action grid by tracking (+0.26 bps), so most of the remaining gap is the 12-action discretisation rather than failed

Table 1: Held-out test results (bps, lower is better). First row: AC objective; other rows: objective minus AC on the same paths. Baselines: 95% CI over paths (real: clustered over 803 dates). Agents: mean ± t-interval over 5 seeds.

Strategy	Exp A random walk	Exp B AR(1) drift	Real NSE 2023–26
AC objective	49.34±1.69	49.25±2.38	55.23±4.31
AC on action grid	+0.26±0.01	+0.27±0.01	+0.30±0.04
TWAP	+7.60±0.61	+7.59±0.90	+8.24±1.18
VWAP	—	+7.23±0.90	+11.84±1.23
Signal rule (heuristic)	—	−9.85±0.80	—
Immediate	+451.2	+451.3	+445.3
DQN, Exp A agent	**+0.14**±0.04	—	**+0.15**±0.10
PPO, Exp A agent	**+0.17**±0.07	—	**+0.17**±0.09
DQN, signal observed	—	**−13.94**±1.16	+8.75±5.38
PPO, signal observed	—	**−15.01**±0.99	+2.22±0.90
DQN, signal hidden	—	+1.07±0.51	+2.22±0.48
PPO, signal hidden	—	+0.70±0.90	+2.58±0.65

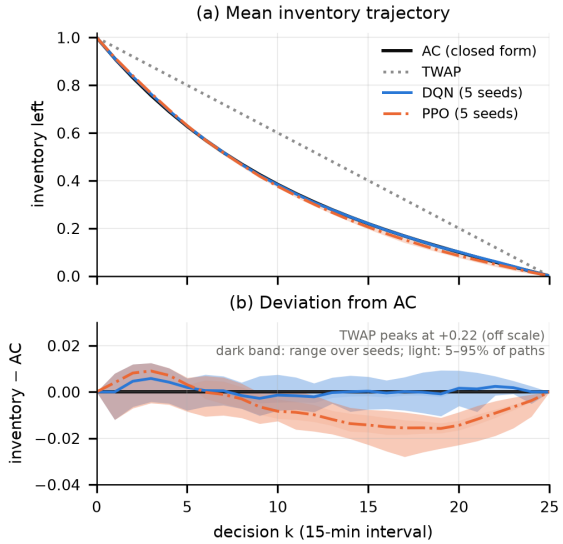


Figure 1: Exp A (test). (a) Mean inventory of DQN and PPO (5 seeds each) vs. AC and TWAP. (b) Deviation from AC; dark band: range over seeds, light: 5–95% of paths.

learning. DQN was slightly closer to AC (trajectory RMSE 0.006 vs. 0.011) though noisier during training.

Experiment B: RL beats the optimum when it can see a signal. With a predictable AR(1) drift, agents that observe it beat AC by 13.9±1.2 (DQN) and 15.0±1.0 bps (PPO), i.e. about 30% of the total cost, and every per-seed CI excludes zero (Fig. 2). They also beat a hand-written rule ("sell slower when drift is positive") by 4–5 bps, so RL learns more than the obvious heuristic. Mechanistically, the agents cut mean IS from 30.7 to 10–13 bps at the price of a higher risk term (21–25 vs. 18.5), while the tail cost improves ($\text{CVaR}_{5\%}$ 242–254 vs. 281 bps). A regression of trade deviation on observed drift gives slopes of −1.70 (DQN) and −1.50 (PPO) vs. −1.41 for the rule, which confirms the agents are trading on the signal. **Ablation.** Retraining with the drift hidden from the state removes the edge entirely: the ablated agents score +1.07 and +0.70 bps relative to AC, and the matched-seed difference is −15.0±1.5 (DQN) and −15.7±1.5 bps (PPO). The last-price-move feature remains visible and correlates weakly (≈0.3) with the drift, yet yields no measurable advantage.

Real NSE data: transfer. On 803 held-out trading days (2,409 stock-days), the signal-free Exp A agents match AC (+0.15/+0.17 bps) and beat TWAP by 8.1 and historical VWAP by 11.7 bps, with the same sign on each stock (Fig. 3). Signal-trained agents do *not* transfer: real days carry no drift feature,

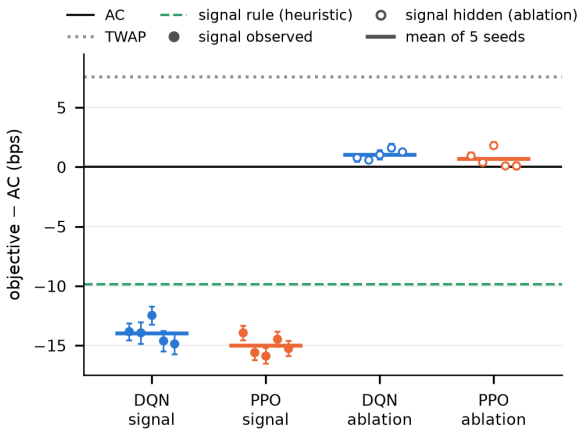


Figure 2: Exp B (test, 10,000 paths): objective minus AC per seed (dots, 95% CI) and mean (bar). Signal observed vs. hidden (ablation); TWAP and the signal rule are reference lines.

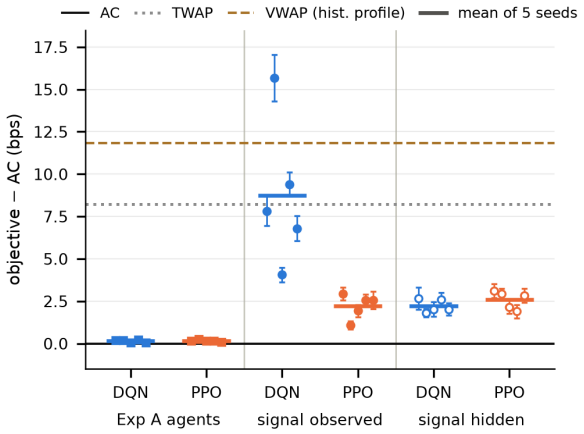


Figure 3: Real NSE test days 2023–26 (simulated impact): objective minus AC for every agent (dots, date-clustered 95% CI) and seed mean (bar).

so they always observe $\mu=0$, a region training never targeted. There PPO loses 2.2 bps and DQN holds too much inventory, losing 8.8 bps on average with a wide seed spread (+4 to +16; two seeds are worse than TWAP). VWAP trails TWAP because in the AC impact model cost depends on trade size but not on market volume, so trading heavily at the open and close is only penalised (a modelling limitation).

5 Discussion and Limitations

(1) Given the right reward, generic RL rediscovers a classical closed-form solution, validating both environment and learners. (2) RL's value lies where closed forms stop: using state information a static schedule cannot. (3) That value needs the signal at deployment; without it the signal-free agent is the safe default. Limitations: impact is simulated with a linear AC model (no order book, no volume-dependent impact), σ is fixed, the synthetic signal is noiseless and correctly specified (an upper bound on real gains), we tested a single λ, only 5 seeds per arm, the ablation still exposes the last price move, and Exp A and Exp B test paths share their noise draws. Extensions: a noisy real signal, volume-dependent impact, a λ sweep.

6 Conclusion

RL matches the Almgren–Chriss optimum where it is optimal, beats it by ∼30% when a predictive signal is observable, and loses the edge exactly when the signal is removed, all under a leakage-free freeze-then-test-once protocol.

References

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